

Projection-Retraction MPPI: Exact Constraint-Manifold Control for Manipulators

Abstract—Two arms that rigidly carry one object form a closed kinematic chain: the relative pose of the grippers must not change, at any instant, or internal forces build up and the load tips. Sampling-based controllers such as Model Predictive Path Integral (MPPI) control encode such equality constraints as soft cost penalties, which a finite weight satisfies only approximately—and only by trading them against the task. We make the constraint a property of the sampler instead. Projection-Retraction MPPI (PR-MPPI) projects each sampled control onto the null space of the constraint Jacobian, so every rollout evolves tangent to the constraint manifold, and returns the single executed configuration to the manifold with a few Gauss–Newton iterations; the MPPI update is otherwise unchanged. Feasibility then never competes with the task: the constraint holds to numerical precision whatever the task weights, and an unreachable target makes the robot stop on the constraint rather than leave it. We prove the projected samples are a consistent importance sampler for the constrained problem, bound the tangent-step drift the retraction removes, and show that retracting only the executed state perturbs the sampler by no more than second order in the time step. The same two operators stay bounded at kinematic singularities through a truncated inverse and steer clear of joint limits through a null-space drift. On a 14-degree-of-freedom dual-arm tray task the constraint holds near 10^{-10} where soft-penalty and projection-plus-gain baselines reach 10^{-1} and 10^{-4} , and a GPU implementation runs well inside its control period; **a hardware demonstration on two manipulators confirms the behaviour.**

Index Terms—Sampling-based model predictive control, model predictive path integral, equality-constrained control, null-space projection, dual-arm manipulation, constrained motion control.

I. INTRODUCTION

MANY manipulation tasks ask the robot to do more than reach a goal: they require it to hold a kinematic equality constraint on its configuration throughout the motion. A wiping or polishing tool must stay on the work surface; a carried tray or glass must stay level; two arms that rigidly hold the same object form a closed kinematic chain whose relative pose must not change. Each case ties the joint configuration to a fixed relationship that must hold *exactly* rather than on average—a closed chain that drifts even slightly builds up internal wrenches or loses the grasp, and a tray that tilts a few degrees spills. The controller must therefore keep the robot on the constraint at every step while it makes progress on the task.

Classical projection-based kinematic control holds such constraints exactly, but optimizes the task only greedily, one step at a time, and stalls on the nonconvex costs that arise with obstacles or multimodal goals. Model Predictive Path Integral (MPPI) control is the opposite: it scores sampled control sequences by cost, needing neither gradients nor convexity, and runs in real time on parallel hardware—but the constraint is normally folded into that same cost as a soft penalty, and a soft

penalty cannot make a hard equality hold. A finite weight only balances constraint violation against task progress, so neither is met exactly; raising the weight to tighten the constraint slows the task and sharpens the cost surface the sampler must explore. Some recent controllers instead project the samples onto the null space of the constraint Jacobian [1], [2], keeping each rollout tangent to the constraint; but a tangent step still leaves the configuration slightly off it, and the residual is either left to accumulate or patched by a gain-tuned feedback correction whose precision depends on that tuning [1], with no analysis of what the projection does to the sampler and with singularities handled only through a cost penalty.

We close this gap: the configurations that satisfy the constraint form a smooth surface in joint space, and we hold the robot on it with two operators built from the constraint Jacobian. Each sampled control is first projected onto the Jacobian’s null space, so the rollouts the controller evaluates stay tangent to that surface; the configuration that is actually executed at each step is then *retracted* back onto it by a few Gauss–Newton iterations, which removes the small tangent-step drift. The task is optimized by an otherwise ordinary MPPI update, now sampling within the tangent directions that keep the constraint satisfied. We call this Projection-Retraction MPPI (PR-MPPI).

Handling the constraint with these two operators rather than with the cost has a few practical consequences. Constraint satisfaction no longer competes with the task: it holds to numerical precision whatever weight the task is given, and when the commanded target is unreachable the robot stops on the constraint instead of trading feasibility for a little more reach. The same two operators also cover two issues that otherwise trouble Jacobian-based control: near a kinematic singularity a truncated inverse [3] keeps the projection and retraction bounded, and joint limits are avoided by a null-space drift [4], [5] that steers the motion away from them while keeping the rollouts tangent. And because every operation batches across samples, the controller ports directly to a GPU and runs in real time.

We make four contributions:

- PR-MPPI, which enforces a hard kinematic equality constraint in MPPI by projecting the samples onto the constraint Jacobian’s null space and retracting the executed configuration onto the constraint, so it holds exactly and independently of the task cost;
- an analysis of the scheme: the projected samples are a consistent importance sampler for the constrained problem, the per-step drift is $O(\Delta t^2)$ and accumulates to at most $O(H\Delta t^2)$ over a rollout, the retraction removes it, and retracting only the executed state stays within

TABLE I

SAMPLING-BASED CONTROLLERS THAT CONSTRAIN THE ROLLOUTS. PR-MPPI IS THE ONLY ONE THAT COMBINES EXACT, LEARNING-FREE SATISFACTION OF A HARD EQUALITY CONSTRAINT WITH SINGULARITY ROBUSTNESS AND ANALYTICAL GUARANTEES. “—” MARKS AN ATTRIBUTE THAT DOES NOT APPLY TO THE METHOD.

Method	Year	System	Hard equality	Exact	Learning-free	Singularity-robust	Guarantees
PRIEST [10]	2023	mobile robot	×	—	✓	—	×
π -MPPI [11]	2025	aerial vehicle	×	—	✓	—	×
MPPI-Tan [12]	2025	manipulators	×	—	✓	—	×
Zhu et al. [1]	2026	dual-arm	✓	×	✓	×	×
MC-MPPI [13]	2026	manipulators	✓	✓	×	×	×
PR-MPPI (ours)	—	manipulators	✓	✓	✓	✓	✓

Hard equality: the method targets a hard equality constraint (rather than inequality/soft costs). *Exact*: the equality holds to numerical precision rather than approximately (e.g. via a tuned feedback gain). *Singularity-robust*: the projection and retraction stay bounded at rank-deficient configurations. *Guarantees*: a stated optimality of the importance sampler and a bound on the off-manifold rollout drift. For MC-MPPI, *Exact* is with respect to its learned manifold rather than the true constraint.

$O(H^2 \Delta t^2 / \lambda)$ in KL divergence of retracting every rollout;

- two extensions within the same construction: a truncated constraint inverse that keeps the operators bounded near kinematic singularities, and a null-space drift that avoids joint limits; and
- an evaluation on a 14-degree-of-freedom dual-arm closed chain, where PR-MPPI holds the constraint at $\sim 10^{-9}$, stays robust at singularities and joint limits, stops on the constraint under unreachable targets, and runs in real time on a GPU, while soft-penalty and projection-plus-gain [1] baselines do not; and a hardware demonstration of the same task on two manipulators.

II. RELATED WORK

Sampling-based MPC can express goal tracking, smoothness, actuation limits, and collision avoidance directly as cost terms, but because these enter as soft penalties it gives no formal guarantee of constraint satisfaction [6]. Strategies that strengthen this range from penalty-based formulations through probabilistic (chance-constraint) and safety-filtering approaches to constraint-aware sampling, trading modelling generality for the strength of the guarantee: penalty terms apply to any black-box cost but only bias the search; chance constraints and safety filters add probabilistic or barrier-based safety [7], [8] at the price of a differentiable model; and constraint-aware sampling shifts infeasible samples toward the feasible set, for instance by a primal–dual correction [9] or a projection optimizer [10]. These families all target *inequality* requirements—collisions, joint and workspace bounds—and deliver approximate or probabilistic feasibility; holding a robot *exactly* on a hard equality constraint is outside their scope.

Closest to our mechanism is a line of work that constrains the samples themselves rather than the cost. PRIEST projects sampled trajectories onto a collision-and-kinematics feasible set by alternating minimization for navigation [10]; π -MPPI projects the control samples onto input- and smoothness-bound sets for fixed-wing flight [11]; and MPPI-Tan restricts manipulator sampling to the tangent space of an obstacle gradient to escape local minima [12]. All three project for inequality or obstacle constraints—the constraint-aware-sampling family above—rather than for a hard equality, and none maintains an exact equality manifold or analyzes the effect of the projection

on the proposal. A related path-integral scheme instead shapes the proposal with a *learned* importance-sampling policy toward feasible, low-cost regions [2], again without an exact equality manifold and with a learned rather than analytic model.

For manipulators that must hold a kinematic *equality*, two recent controllers are most related. The dual-arm controller of Zhu et al. [1] enforces a closed chain by projecting joint samples onto the constraint Jacobian’s null space and corrects the residual tangent drift with a low-level feedback gain; it encodes the tray-level condition as a soft tilt cost and steers away from singular configurations through a manipulability cost rather than handling them in the operators themselves. MC-MPPI [13] instead learns the feasible set as a variational-autoencoder latent manifold and retracts onto it with a quadratic program. We keep the same null-space projection but make the return step an analytic retraction whose precision does not depend on a feedback gain, and we treat orientation and level as an equality on the manifold rather than a soft cost. The same operators stay bounded at singularities, and the method uses no learned model.

The projection-and-retraction recipe itself is classical in constrained sampling-based *planning*, where samples are projected or retracted onto an implicitly defined constraint manifold [14]–[16]. Those methods build a path offline under the constraint; we bring the same two operators into the real-time MPC loop, where the projector reshapes the proposal distribution itself, and analyze what this does to the sampler and to the executed trajectory.

In short (Table I), existing sampling-based controllers either keep constraints in the cost, or project for inequality constraints, or plan offline on the manifold, or—when they hold an equality online—rely on a tuned feedback correction or a learned manifold; none returns the executed state to a hard equality by an exact, analyzed step while also staying robust to the singularities and joint limits that redundant arms encounter. PR-MPPI is our response to this gap: an analytic projection-and-retraction that holds a hard equality constraint to numerical precision with a stated optimality and drift analysis, and that extends to singularities and joint limits within the same construction.

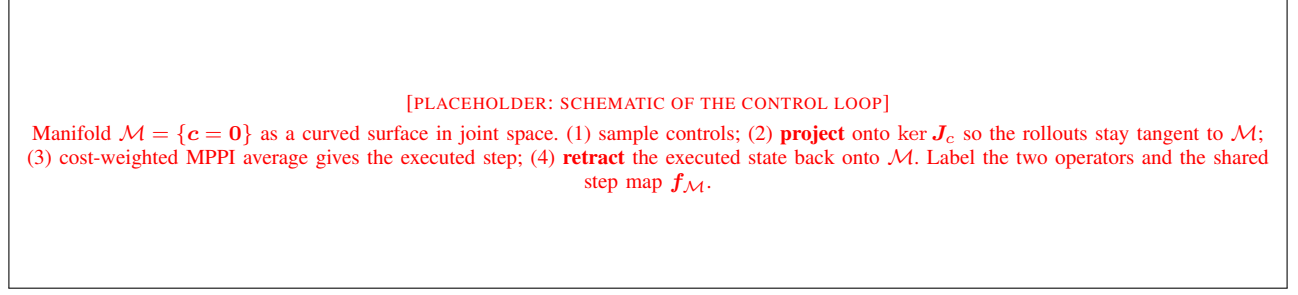


Fig. 1. **Projection-Retraction MPPI.** [TODO: Each sampled control is projected onto the null space of the constraint Jacobian, so the rollouts stay tangent to the constraint manifold $\mathcal{M}$; the cost-weighted MPPI update selects a tangent step; the executed configuration is retracted back onto $\mathcal{M}$.]

III. PROJECTION-RETRACTION MPPI

PR-MPPI enforces the equality constraint with two analytic operators built from the constraint Jacobian: a null-space *projection* that confines the sampling to the tangent space of $\mathcal{M}$, and a *retraction* that returns the executed configuration to $\mathcal{M}$ (Fig. 1). We introduce both in a single constrained control step and give the full algorithm there, so the method can be implemented from one place. We then justify the scheme in one analysis: the projection is the constrained-optimal proposal (Lemma 1), the retraction makes the executed trajectory exact and independent of tuning (Proposition 1), and retracting only the executed state—rather than every rollout—changes the sampler only at second order in the step size (Proposition 2). We close with the variants that keep the operators bounded near singularities and steer the sampling clear of joint limits.

A. Problem Setup

We control a manipulator with joint configuration $\mathbf{q} \in \mathbb{R}^n$ through joint-velocity commands $\mathbf{u} \in \mathbb{R}^n$, under the kinematic model

$$\mathbf{q}_{t+1} = \mathbf{q}_t + \Delta t \mathbf{u}_t. \quad (1)$$

The task must respect m holonomic equality constraints $\mathbf{c}(\mathbf{q}) = \mathbf{0}$, with $\mathbf{c} : \mathbb{R}^n \rightarrow \mathbb{R}^m$ twice continuously differentiable; their zero set

$$\mathcal{M} = \{\mathbf{q} \in \mathbb{R}^n : \mathbf{c}(\mathbf{q}) = \mathbf{0}\}, \quad \mathbf{J}_c(\mathbf{q}) = \frac{\partial \mathbf{c}}{\partial \mathbf{q}} \in \mathbb{R}^{m \times n}, \quad (2)$$

is the feasible manifold and $\mathbf{J}_c$ its constraint Jacobian, of full row rank $m < n$ away from the isolated singular configurations treated in Sec. III-D. The running instance in this paper is a dual-arm closed chain: two 7-DoF arms rigidly hold a tray ($n = 14$), and $\mathbf{c}$ stacks the grippers' relative position with the level condition on both end-effector axes ($m = 7$). We seek to minimize an expected horizon- H task cost while keeping *every* executed configuration on $\mathcal{M}$,

$$\min_{\mathbf{U}} \mathbb{E} \left[\phi(\mathbf{q}_H) + \sum_{t=0}^{H-1} \ell(\mathbf{q}_t, \mathbf{u}_t) \right] \quad \text{s.t.} \quad \mathbf{q}_t \in \mathcal{M} \quad \forall t, \quad (3)$$

with $\mathbf{U} = (\mathbf{u}_0, \dots, \mathbf{u}_{H-1})$ the nominal control sequence, ϕ a terminal and ℓ a running cost.

MPPI solves the *unconstrained* version of (3) by sampling. Around the nominal sequence it draws K rollouts with $\mathbf{u}_t^k = \mathbf{u}_t + \boldsymbol{\varepsilon}_t^k$, $\boldsymbol{\varepsilon}_t^k \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma})$, scores each by its rollout cost $S^k =$

$\phi(\mathbf{q}_H^k) + \sum_t \ell(\mathbf{q}_t^k, \mathbf{u}_t^k)$, and updates the nominal sequence by the cost-weighted average

$$\mathbf{u}_t \leftarrow \mathbf{u}_t + \sum_{k=1}^K w^k \boldsymbol{\varepsilon}_t^k, \quad w^k = \frac{\exp(-S^k/\lambda)}{\sum_j \exp(-S^j/\lambda)}, \quad (4)$$

with temperature $\lambda > 0$, after which $\mathbf{u}_0$ is executed and the horizon recedes [6], [17]. The update is gradient-free and parallel, but nothing in (4) keeps the rollouts or the executed state on $\mathcal{M}$; writing $\mathbf{c} = \mathbf{0}$ as a penalty in ℓ only trades violation against task progress. The next section confines the sampling to $\mathcal{M}$ and restores exact feasibility with two operators.

B. The Constrained Control Step

Both operators are built from the constraint Jacobian. At a configuration $\mathbf{q}$ we form the orthogonal null-space projector of $\mathbf{J}_c$, the standard tool of redundancy resolution [18],

$$\mathbf{N}(\mathbf{q}) = \mathbf{I} - \mathbf{J}_c^\#(\mathbf{q}) \mathbf{J}_c(\mathbf{q}), \quad \mathbf{J}_c^\# = \mathbf{J}_c^\top (\mathbf{J}_c \mathbf{J}_c^\top)^{-1}, \quad (5)$$

and propagate every motion—sampled or executed—through the *constrained step map*

$$\mathbf{f}_{\mathcal{M}}(\mathbf{q}, \mathbf{v}) = \mathbf{q} + \Delta t \mathbf{N}(\mathbf{q}) \mathbf{v}. \quad (6)$$

Rollouts are generated by sending the *total* commanded velocity through (6),

$$\mathbf{q}_{t+1}^k = \mathbf{f}_{\mathcal{M}}(\mathbf{q}_t^k, \mathbf{u}_t + \boldsymbol{\varepsilon}_t^k), \quad \boldsymbol{\varepsilon}_t^k \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}). \quad (7)$$

Projecting $\mathbf{u}_t + \boldsymbol{\varepsilon}_t^k$ as a whole—not the noise alone—matters: the nominal $\mathbf{u}_t$ was computed at a different configuration and need not be tangent at $\mathbf{q}_t^k$, and it makes the rollouts use exactly the dynamics through which the optimized control is later executed. By construction $\mathbf{J}_c \mathbf{N} = \mathbf{0}$, so the constraint is preserved to first order along each rollout, and the effective perturbation $\mathbf{N} \boldsymbol{\varepsilon}_t^k$ has the degenerate law $\mathcal{N}(\mathbf{0}, \mathbf{N} \boldsymbol{\Sigma} \mathbf{N})$ supported on the tangent space. The MPPI update (4) runs in the unprojected control coordinates—the nominal $\mathbf{U}$ is stored unprojected and the projection applied only at the point of use—so the same nominal stays valid as the configuration evolves.

A tangent step preserves the constraint only to first order, so over a finite step the configuration drifts off the curved manifold by $O(\Delta t^2)$ (Proposition 1). We close this gap with a *retraction* applied to the single executed state each control

Algorithm 1 Projection-Retraction MPPI (one control step)

Require: current $\mathbf{q}$ on $\mathcal{M}$ (else $\mathbf{q} \leftarrow \mathcal{R}_{\mathcal{M}}(\mathbf{q})$), nominal $\mathbf{U}$, samples K , horizon H

- 1: **for** $k = 1$ **to** K **do**
- 2: $\mathbf{q}_0^k \leftarrow \mathbf{q}$; sample $\boldsymbol{\varepsilon}_{0:H-1}^k \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma})$
- 3: **for** $t = 0$ **to** $H - 1$ **do**
- 4: $\mathbf{q}_{t+1}^k \leftarrow \mathbf{f}_{\mathcal{M}}(\mathbf{q}_t^k, \mathbf{u}_t + \boldsymbol{\varepsilon}_t^k)$; accumulate S^k // Eq. (6)
- 5: **end for**
- 6: **end for**
- 7: $\mathbf{U} \leftarrow \mathbf{U} + \sum_k w^k \boldsymbol{\varepsilon}^k$, weights from Eq. (4)
- 8: execute: $\mathbf{q} \leftarrow \mathbf{f}_{\mathcal{M}}(\mathbf{q}, \mathbf{u}_0)$
- 9: retract: $\mathbf{q} \leftarrow \mathcal{R}_{\mathcal{M}}(\mathbf{q})$ // Eq. (8)
- 10: shift $\mathbf{U}$ (receding horizon)

step: from the post-step configuration we iterate the Gauss–Newton update

$$\mathbf{q} \leftarrow \mathbf{q} - \mathbf{J}_c^\#(\mathbf{q}) \mathbf{c}(\mathbf{q}) \quad \text{until} \quad \|\mathbf{P}_r \mathbf{c}(\mathbf{q})\| < \epsilon, \quad (8)$$

where $\mathbf{P}_r$ projects onto the *retained* constraint directions—equal to $\mathbf{I}$ unless a direction is truncated near a singularity (Sec. III-D)—so the iteration stops on the residual it can actually correct. Because it starts $O(\Delta t^2)$ -close to $\mathcal{M}$ where $\mathbf{J}_c$ has full row rank, (8) converges locally at a quadratic rate—two or three iterations drive that residual to numerical zero—so the executed trajectory lies on $\mathcal{M}$ to machine precision (up to any truncated direction). The retraction touches only the executed state, not the $K \times H$ rollouts—a design choice Proposition 2 justifies.

Algorithm 1 collects the step. The sampled rollouts and the executed control share the one step map $\mathbf{f}_{\mathcal{M}}$, so the optimized control is applied exactly as it was evaluated; the two robustness variants below change only how $\mathbf{f}_{\mathcal{M}}$ is formed, by replacing the inverse $\mathbf{J}_c^\#$ (Sec. III-D) or adding a null-space drift (Sec. III-E).

C. Analysis

Three questions decide whether the scheme is principled: is the projected proposal the right one; is the executed trajectory exactly feasible; and does retracting only the executed state—rather than every rollout—lose anything? We answer each in turn.

The projection is the right proposal: Recall the free-energy view of MPPI: with base perturbation $p(\delta \mathbf{q}) = \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma})$ at the current configuration $\mathbf{q}$, the distribution minimizing $\mathbb{E}_\rho[S] + \lambda \text{KL}(\rho \| p)$ is

$$\rho^*(\delta \mathbf{q}) \propto p(\delta \mathbf{q}) \exp(-S(\mathbf{q} + \delta \mathbf{q})/\lambda), \quad (9)$$

and the MPPI update is the self-normalized importance-sampling (SNIS) estimate of its mean. Imposing the constraint confines the admissible perturbations to the tangent space $T = \ker \mathbf{J}_c(\mathbf{q})$. We take the base measure on T to be the projection (pushforward) of p through $\mathbf{N}$, namely $p_T = \mathcal{N}(\mathbf{0}, \mathbf{N}\boldsymbol{\Sigma}\mathbf{N})$ —the measure the sampler actually realizes—and define the

constrained-optimal distribution as the corresponding tilt of (9),

$$\rho_{\mathcal{M}}^*(\delta \mathbf{q}) \propto p_T(\delta \mathbf{q}) \exp(-S(\mathbf{q} + \delta \mathbf{q})/\lambda), \quad \delta \mathbf{q} \in T, \quad (10)$$

densities and the KL divergence taken on the $(n - m)$ -dimensional subspace T . For isotropic $\boldsymbol{\Sigma}$ this pushforward coincides with conditioning p on T ; for general $\boldsymbol{\Sigma}$ it is a different, projected Gaussian.

Lemma 1 (Constrained-optimal proposal). *Fix $\mathbf{q} \in \mathcal{M}$ and let $T = \ker \mathbf{J}_c(\mathbf{q})$. Draw $\boldsymbol{\varepsilon}_k \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma})$, set $\delta \mathbf{q}_k = \mathbf{N}\boldsymbol{\varepsilon}_k$, and weight by (4). Then $\sum_k w_k \delta \mathbf{q}_k$ is a consistent (as $K \rightarrow \infty$) SNIS estimator of $\mathbb{E}_{\rho_{\mathcal{M}}^*}[\delta \mathbf{q}]$, so the tangent step realized by the updated nominal in Algorithm 1 matches the first moment of $\rho_{\mathcal{M}}^*$ in the large-sample limit.*

Proof. Since $\mathbf{N}$ is the orthogonal projector onto T , the projected samples are i.i.d. with law exactly p_T on T , and the weights $w_k \propto \exp(-S/\lambda)$ are precisely the self-normalized importance weights for the change of measure from p_T to $\rho_{\mathcal{M}}^*$, whose Radon–Nikodym derivative is proportional to $\exp(-S/\lambda)$. SNIS consistency gives the claim; since $\mathbf{N}$ is fixed at $\mathbf{q}$ and linear, $\sum_k w_k \delta \mathbf{q}_k = \mathbf{N} \sum_k w_k \boldsymbol{\varepsilon}_k$, the projected image of (4) through (6). $\square$

The retraction makes it exact: The tangent restriction enforces the constraint only to first order; the retraction closes the gap.

Proposition 1 (Rollout drift and executed feasibility). *Suppose $\mathbf{c}$ is twice continuously differentiable with $\|\nabla^2 c_i(\mathbf{q})\| \leq L_c$ for all i on a neighborhood of $\mathcal{M}$, and let the commanded velocities be bounded, $\|\mathbf{u}_t + \boldsymbol{\varepsilon}_t^k\| \leq v_{\max}$. Then: (i) each constrained step (6) satisfies*

$$\|\mathbf{c}(\mathbf{f}_{\mathcal{M}}(\mathbf{q}, \mathbf{v}))\| \leq \|\mathbf{c}(\mathbf{q})\| + \frac{\sqrt{m} L_c}{2} \Delta t^2 \|\mathbf{v}\|^2; \quad (11)$$

(ii) a rollout of length $t \leq H$ from $\mathbf{q}_0^k \in \mathcal{M}$ obeys $\|\mathbf{c}(\mathbf{q}_t^k)\| \leq \frac{\sqrt{m} L_c}{2} v_{\max}^2 t \Delta t^2 = O(H \Delta t^2)$; and (iii) for sufficiently small Δt , the executed state after the retraction (8) satisfies $\|\mathbf{c}(\mathbf{q})\| < \epsilon$ at every control step, independently of the task cost ℓ , ϕ , the temperature λ , and the sampling covariance $\boldsymbol{\Sigma}$.

Proof. (i) A Taylor expansion of c_i about $\mathbf{q}$ has its linear term vanish because $\mathbf{J}_c \mathbf{N} = \mathbf{0}$ and $\|\mathbf{N}\mathbf{v}\| \leq \|\mathbf{v}\|$; the second-order remainder is bounded by $\frac{L_c}{2} \Delta t^2 \|\mathbf{v}\|^2$ per component, and stacking the m components yields (11). (ii) telescopes (11) from $\mathbf{c}(\mathbf{q}_0^k) = \mathbf{0}$. (iii) is the stopping criterion of (8), which by (ii) starts $O(\Delta t^2)$ -close to $\mathcal{M}$, inside the local quadratic-convergence basin of Gauss–Newton for full-row-rank $\mathbf{J}_c$; none of its quantities depend on ℓ , ϕ , λ , or $\boldsymbol{\Sigma}$. $\square$

Together the two results make the decoupling precise: rollout costs are evaluated at most $O(H \Delta t^2)$ off the manifold—vanishing with the control rate, a scaling the experiments confirm directly (Sec. IV)—while the *executed* trajectory satisfies $\mathbf{c}(\mathbf{q}) = \mathbf{0}$ to numerical precision regardless of how the task is weighted or tempered. A soft penalty, by contrast, has a stationary violation set by its weight that never vanishes.

Retracting only the executed state is enough: Algorithm 1 evaluates the rollout costs at the tangent states of (7), which sit $O(H\Delta t^2)$ off $\mathcal{M}$. The alternative retracts *every* rollout state,

$$\tilde{\mathbf{q}}_{t+1}^k = \mathcal{R}_{\mathcal{M}}(\mathbf{f}_{\mathcal{M}}(\tilde{\mathbf{q}}_t^k, \mathbf{u}_t + \boldsymbol{\varepsilon}_t^k)), \quad (12)$$

so that costs are evaluated exactly on $\mathcal{M}$. Both schemes are tilts of the same base measure $p(\boldsymbol{\varepsilon}) = \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma})^{\otimes H}$ through their rollout costs $\tilde{S}, \bar{S}$:

$$\tilde{\rho} \propto p e^{-\tilde{S}/\lambda}, \quad \bar{\rho} \propto p e^{-\bar{S}/\lambda}, \quad (13)$$

with $\bar{\rho}$ the path-level on-manifold proposal—the horizon- H analogue of (10).

Proposition 2 (Executed-only vs. per-rollout retraction). *Assume, on a neighborhood of $\mathcal{M}$ containing both rollout families: $\sigma_{\min}(\mathbf{J}_c) \geq \underline{\sigma} > \mu$, $\|\nabla^2 c_i\| \leq L_c$, $\mathbf{N}(\cdot)$ is L_N -Lipschitz, $\ell(\cdot, \mathbf{u})$ and ϕ are L_S -Lipschitz in $\mathbf{q}$, and $\|\mathbf{u}_t + \boldsymbol{\varepsilon}_t^k\| \leq v_{\max}$. Then for sufficiently small Δt , with $T = H\Delta t$: (i) under the same noise,*

$$\begin{aligned} \|\tilde{\mathbf{q}}_t^k - \bar{\mathbf{q}}_t^k\| &\leq C_0 t \Delta t^2 e^{L_N v_{\max} T}, \\ |\tilde{S}^k - \bar{S}^k| &\leq \delta := C_1 H^2 \Delta t^2, \end{aligned} \quad (14)$$

with $C_0 = \sqrt{m} L_c v_{\max}^2 / \underline{\sigma}$ and $C_1 = L_S C_0 e^{L_N v_{\max} T}$; (ii) the tilts (13) satisfy $e^{-2\delta/\lambda} \leq d\bar{\rho}/d\tilde{\rho} \leq e^{2\delta/\lambda}$, hence $\text{KL}(\bar{\rho} \parallel \tilde{\rho}) \leq 2\delta/\lambda = O(H^2 \Delta t^2 / \lambda)$ and $\text{TV}(\bar{\rho}, \tilde{\rho}) \leq \sqrt{\delta/\lambda}$; and (iii) the MPPI updates differ by $\|\mathbb{E}_{\tilde{\rho}}[\boldsymbol{\varepsilon}_t] - \mathbb{E}_{\bar{\rho}}[\boldsymbol{\varepsilon}_t]\| \leq (e^{2\delta/\lambda} - 1) \mathbb{E}_{\tilde{\rho}}[\|\boldsymbol{\varepsilon}_t\|] = O(H^2 \Delta t^2 / \lambda)$, so for fixed H the executed-only scheme converges to the fully on-manifold scheme as $\Delta t \rightarrow 0$.

Proof sketch. (i) One step splits into the retraction displacement—at most $(2/\underline{\sigma})\|\mathbf{c}(\mathbf{q}^+)\| \leq C_0 \Delta t^2$ by (11) and quadratic local convergence (taking the tolerance $\epsilon = 0$ for clarity; otherwise an additive $O(\epsilon)$ enters (14))—and a Lipschitz propagation factor $(1 + L_N v_{\max} \Delta t)$; the discrete Grönwall inequality gives the state bound, and summing the L_S -Lipschitz costs over the horizon gives δ . (ii) For two tilts of one base measure with $|\tilde{S} - \bar{S}| \leq \delta$ pointwise, the log density ratio (including normalizers) is bounded by $2\delta/\lambda$, which gives the sandwich, the KL bound, and via Pinsker the TV bound. (iii) Apply the density-ratio sandwich to $\mathbf{f} = \boldsymbol{\varepsilon}_t$. $\square$

Remark 1. Lemma 1 concerns the proposal mean (the covariance stays $\mathbf{N}\boldsymbol{\Sigma}\mathbf{N}$, not adapted) and is exact at $t = 0$, where $\mathbf{N}(\mathbf{q})$ is common to all rollouts; for $t > 0$ the scheme is standard MPPI under the modified dynamics $\mathbf{f}_{\mathcal{M}}$, and Proposition 2 closes the gap over the whole horizon: the realized proposal is within $\text{KL} = O(H^2 \Delta t^2 / \lambda)$ of the on-manifold ideal. Per-rollout retraction would buy that $O(\Delta t^2)$ correction at $(1 + N_{\text{GN}}) \approx 3$ times the rollout cost (Sec. III-F). The Lipschitz-cost assumption asks the obstacle terms in ℓ to be smooth (e.g. built on signed distances).

D. Singularities: Keeping the Operators Bounded

Near a rank deficiency of $\mathbf{J}_c$, a vanishing singular value makes the inverse in (5) and (8) ill-conditioned. Writing the singular value decomposition $\mathbf{J}_c = \mathbf{U}\boldsymbol{\Sigma}_c\mathbf{V}^\top$ with $\boldsymbol{\Sigma}_c =$

$\text{diag}(\sigma_1, \dots, \sigma_m)$, we replace the pseudo-inverse by the truncated inverse

$$\mathbf{J}_c^\# = \mathbf{V}\boldsymbol{\Sigma}_c^\# \mathbf{U}^\top, \quad [\boldsymbol{\Sigma}_c^\#]_{ii} = \frac{h(\sigma_i)}{\sigma_i}, \quad (15)$$

with the *hard* activation $h(\sigma) = 1$ for $\sigma > \mu$ and $h(\sigma) = 0$ otherwise, truncation scale $\mu > 0$: a direction with $\sigma_i \leq \mu$ is dropped from both $\mathbf{N}$ and the retraction, so the controller neither explores nor corrects along it. The retraction's stopping test (8) accordingly uses the retained-direction projector $\mathbf{P}_r = \mathbf{U} \text{diag}(h(\sigma_i)) \mathbf{U}^\top$, terminating on the correctable residual with a fixed cap of N_{GN} iterations as a guard. Because $h \in \{0, 1\}$, the projector $\mathbf{N}$ stays *exact*—an orthogonal projector onto the retained tangent subspace—so whenever every $\sigma_i > \mu$ (the generic case away from singularities) the operators of Sec. III-B are recovered exactly and the analysis of Sec. III-C holds verbatim.

Proposition 3 (Bounded operators under truncation). *With the hard truncation (15), for any configuration $\mathbf{q}$ (including rank-deficient ones): (i) $\mathbf{J}_c^\# \mathbf{J}_c$ is symmetric and idempotent with eigenvalues in $\{0, 1\}$, so $\mathbf{N} = \mathbf{I} - \mathbf{J}_c^\# \mathbf{J}_c$ is the orthogonal projector onto the retained tangent subspace and $\|\mathbf{N}\| \leq 1$; (ii) $\|\mathbf{J}_c^\#\| = \max_{\sigma_i > \mu} 1/\sigma_i \leq 1/\mu$; hence (iii) every commanded velocity satisfies $\|\mathbf{N}\mathbf{v}\| \leq \|\mathbf{v}\|$ and every retraction step satisfies $\|\mathbf{J}_c^\# \mathbf{c}\| \leq \|\mathbf{c}\|/\mu$, uniformly in the singular values.*

Proof. $\mathbf{J}_c^\# \mathbf{J}_c = \mathbf{V} \text{diag}(h(\sigma_i)) \mathbf{V}^\top$ with $h \in \{0, 1\}$ is symmetric and idempotent, which gives (i); on the retained directions $[\boldsymbol{\Sigma}_c^\#]_{ii} = 1/\sigma_i \leq 1/\mu$, which gives (ii); (iii) follows by submultiplicativity. $\square$

Remark 2. The price is that the constraint residual along a truncated direction is left uncorrected—unavoidable once actuation there is lost—in exchange for bounded joint velocities; the experiments report any such residual. To avoid chattering as a singular value crosses the threshold, truncation is engaged and released with hysteresis ($\mu_{\text{off}} < \mu_{\text{on}}$).

E. Joint and Velocity Limits

Joint limits live in the redundancy of the constraint, so they can be respected without touching feasibility. From a barrier $H(\mathbf{q})$ that grows as a joint nears a limit, we add the avoidance velocity $-\nabla H(\mathbf{q})$ to the commanded velocity before the projection, so every rollout and the executed step carry its tangent component $-\mathbf{N}\nabla H$: the redundant motion is steered away from the limits while $\mathbf{J}_c \mathbf{N} = \mathbf{0}$ leaves the constraint untouched—the closed-loop inverse-kinematics avoidance of [5] inside the sampler. Hard *velocity* limits are imposed by scaling the commanded velocity uniformly, $\mathbf{v} \leftarrow \alpha \mathbf{v}$ with $\alpha \in (0, 1]$, which keeps $\mathbf{N}\mathbf{v}$ in the tangent space and so preserves feasibility, whereas clamping the projected velocity per joint would push the motion off the tangent space—a failure mode the experiments exhibit directly.

Remark 3. A classical alternative reshapes the *metric* instead: the weighted least-norm inverse $\mathbf{J}_c^{\#W} = \mathbf{W}^{-1} \mathbf{J}_c^\top (\mathbf{J}_c \mathbf{W}^{-1} \mathbf{J}_c^\top)^{-1}$ with limit-dependent weights [4], [5]

also preserves $\mathbf{J}_c \mathbf{N}_W = \mathbf{0}$. It redistributes only the least-norm *resolution* of the commanded velocity, however, while the sampled exploration that pushes joints toward their limits is left untouched, so we use the explicit drift.

F. Implementation and Cost

Each rollout step forms $\mathbf{N}$ from $\mathbf{J}_c^\#$, dominated by the symmetric $m \times m$ eigendecomposition of $\mathbf{J}_c \mathbf{J}_c^\top$ —a Cholesky solve when no direction is truncated, and otherwise enough to obtain the singular values and vectors needed for (15) without a full SVD of $\mathbf{J}_c$ —at cost $O(m^2 n + m^3)$; the $K \times H$ rollouts therefore cost $O(KH(m^2 n + m^3))$ and parallelize across samples exactly as standard MPPI. The retraction adds two to three Gauss–Newton iterations on the single executed state; retracting every rollout state instead, as in (12), would multiply the per-step rollout cost by $(1 + N_{\text{GN}}) \approx 3$ while improving the proposal only at order Δt^2 (Proposition 2). Since $m \leq n/2$ in our tasks ($m = 7$, $n = 14$ for the dual-arm tray), the projection is a moderate per-sample overhead over an unconstrained MPPI of the same K and H , and because every operation batches across samples the controller ports directly to a GPU; Sec. IV-D reports the measured rates.

IV. EXPERIMENTS

We evaluate PR-MPPI on one platform throughout: a dual-arm closed chain of two 7-DoF arms ($n = 14$) that rigidly hold a tray and must keep it rigid and level ($m = 7$: the grippers’ relative position together with both end-effector axes vertical) while transporting it. We compare constraint handling head-to-head (Sec. IV-B), exercise the robustness variants (Sec. IV-C), measure the real-time GPU implementation (Sec. IV-D), and **demonstrate the same task on hardware** (Sec. IV-E). Full parameters, supplementary videos, and code are available at the project page.¹

A. Setup

All controllers share the sampler, horizon, and task cost, and differ only in how the constraint is handled. The analytic constraint Jacobian matches finite differences to 10^{-10} , and the common parameters are $K = 240$ rollouts, horizon $H = 18$, step $\Delta t = 0.05$ s, temperature $\lambda = 0.4$, and sampling standard deviation $\sigma = 0.22$ [TODO: confirm final values]. Per task we report the constraint violation $\|\mathbf{c}\|$ (mean and max along the trajectory), the tray task error, the executed-control smoothness ($\frac{1}{T} \sum_t \|\mathbf{u}_t - \mathbf{u}_{t-1}\|$), the joint-limit margin where relevant, and the achieved control rate; lower is better except for the control rate. We compare against (i) *soft penalty*, which adds $w_c \|\mathbf{c}\|^2$ to ℓ , with w_c swept; (ii) *projection + gain*, in the style of [1], which projects the samples but corrects the residual drift with a low-level proportional gain instead of a retraction; and (iii) *ablations* of PR-MPPI—projection-only and per-sample retraction. Greedy, projection-based control stalls in the local minima that arise on nonconvex costs with obstacles or multimodal goals [12], [19]; sampling is what lets PR-MPPI escape them.

¹<https://prmpipi.github.io>

TABLE II
DUAL-ARM TRAY, $N = 24$ RANDOM TARGETS (MEAN $\pm$ STD); THE PROJECTION+GAIN ENTRY IS ITS BEST OVER A GAIN SWEEP, AND $\|\mathbf{c}\|$ IS THE PER-RUN MAXIMUM ALONG THE TRAJECTORY.

Method	$\ \mathbf{c}\ $	tray reach error	smoothness
Soft penalty	$1.3 \times 10^{-1} \pm 1.6 \times 10^{-2}$	0.039 ± 0.046	0.114
Projection + gain [1]	$6.4 \times 10^{-4} \pm 6.0 \times 10^{-4}$	0.014 ± 0.047	0.030
PR-MPPI (ours)	$7.8 \times 10^{-11} \pm 2.1 \times 10^{-11}$	0.014 ± 0.047	0.030

B. Constraint Handling Compared

Table II reports means over $N = 24$ random feasible tray targets: PR-MPPI holds the constraint at machine precision in *every* trial while the soft baseline never does, and the projection+gain baseline sits in between, at a precision set by its gain. Three further experiments isolate *why* (Fig. 2).

Decoupling. Sweeping the soft weight w_c traces a $\|\mathbf{c}\|$ –task-error Pareto front: small weights track well but violate the constraint at the 10^{-1} level of Table II, large weights tighten it only to $\sim 10^{-2}$ while degrading tracking. PR-MPPI sits off this front— $\|\mathbf{c}\| \sim 10^{-10}$ at the same task error—and holding the task fixed while varying λ and the task weight leaves its $\|\mathbf{c}\|$ unchanged, the tuning-independence of Prop. 1(iii) (Fig. 2b). [TODO: run the dual-arm w_c and λ /weight sweeps.]

Versus projection + gain. The gain baseline must be tuned: we expect its $\|\mathbf{c}\|$ to range over several orders of magnitude across the feedback gain k_p (its best in Table II is 6.4×10^{-4}), whereas PR-MPPI holds a flat $\sim 10^{-10}$ for any setting—the retraction’s precision does not depend on tuning (Fig. 2c). [TODO: run the dual-arm k_p sweep.]

Drift scaling. With the retraction disabled (projection-only ablation), the per-step constraint drift should scale as Δt^2 (log–log slope 2), matching Prop. 1(i,ii); enabling the retraction collapses it to the tolerance ϵ (Fig. 2d). [TODO: run the dual-arm Δt sweep.]

C. Robustness: Singularities, Joint Limits, Infeasible Targets

Singularity transit. Steering the 14-DoF configuration toward a rank deficiency of $\mathbf{J}_c$ (smallest singular value $\sigma_{\min} \rightarrow 5 \times 10^{-5}$), the truncated inverse holds $\|\mathbf{J}_c^\#\|$ below $1/\mu = 50$ ($\mu = 2 \times 10^{-2}$), whereas the untruncated pseudo-inverse $1/\sigma_{\min}$ reaches 1.9×10^4 and diverges; the uncorrected residual is confined to the truncated direction, as Proposition 3 predicts.

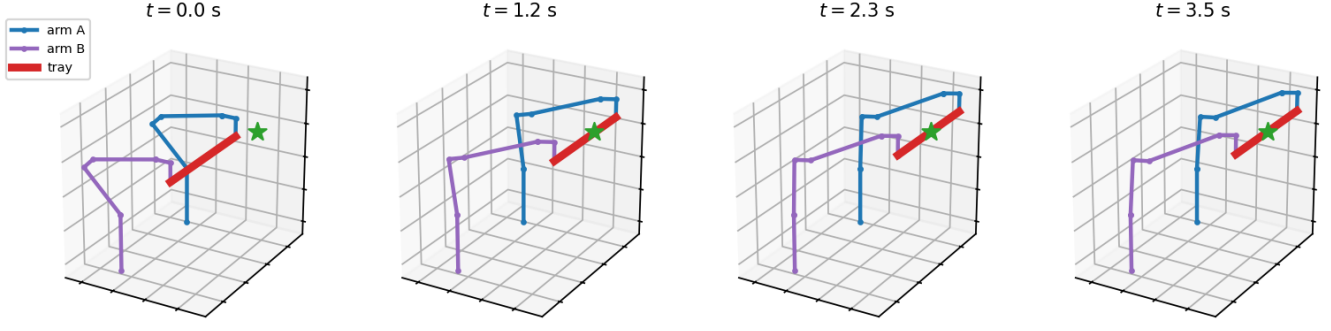
Joint limits. With the null-space avoidance drift of Sec. III-E, every joint stays inside its bound (margin ≥ 0) while $\|\mathbf{c}\|$ holds at machine precision, whereas the plain orthogonal projection drives joints past their limits (Fig. 3a). Restoring the limit instead by clamping the projected velocity breaks tangency and spikes $\|\mathbf{c}\|$ to $\sim 10^{-2}$.

Infeasible target. Commanded to a tray target outside the reachable workspace, PR-MPPI gives up task progress while holding the constraint exactly and keeping joint velocities bounded—the “stops on the constraint” behaviour of Sec. III-C (Fig. 3b).

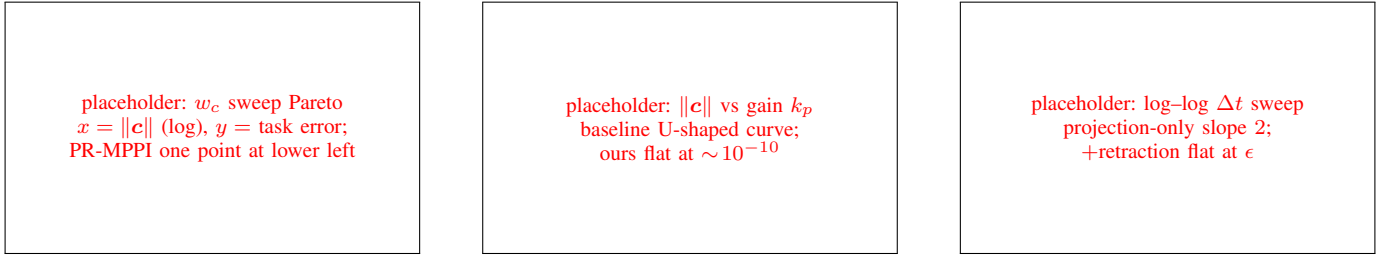
D. Real-Time GPU Implementation

The rollout—the 99% bottleneck of a CPU implementation—batches across samples, so we port the

two 7-DoF arms hold a tray (red) rigid and level while transporting it to the target ★ (14 DoF, $m = 7$ constraint)



(a) Task: two 7-DoF arms hold the tray (red) rigid and level ($m=7$) and transport it to the target; snapshots over the run.

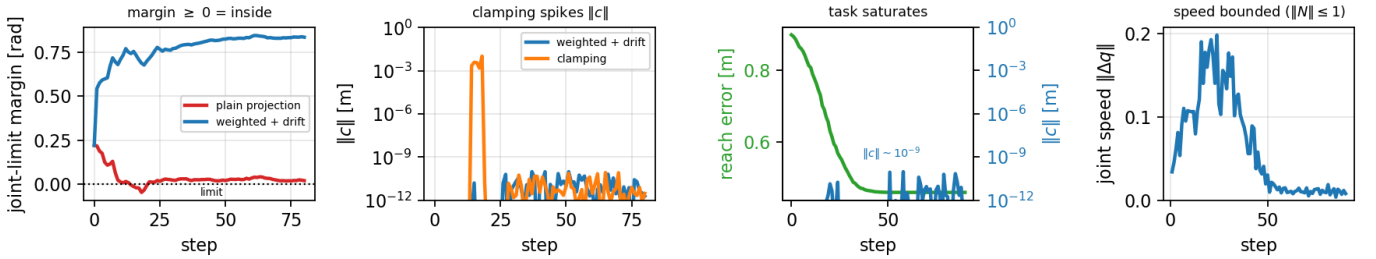


(b) Decoupling: the soft w_c sweep traces a $\|c\|$ -task-error Pareto front; PR-MPPI sits off it.

(c) Vs. projection+gain: the baseline plateaus at a gain-dependent level; PR-MPPI is flat.

(d) Drift: projection-only scales as Δt^2 ; the retraction collapses it.

Fig. 2. **Constraint handling on the dual-arm tray** (14-DoF, $m = 7$): one transport task, three isolated properties. [TODO: render (b)–(d) from the dual-arm sweeps.]



(a) Joint limits: the null-space drift keeps every joint inside its bound (margin ≥ 0 , left) at $\|c\| \sim 10^{-10}$, while plain projection crosses; clamping holds the limit but spikes $\|c\|$ (right).

(b) Infeasible target: the reach error saturates at the workspace boundary while $\|c\|$ stays at $\sim 10^{-10}$ and the joint speed stays bounded.

Fig. 3. **Robustness on the dual-arm tray** (14-DoF, $m = 7$).

controller to JAX and run it on a consumer GPU (NVIDIA RTX 4060). The fused control step (sampling, the projected $K \times H$ rollout with reverse-mode constraint Jacobians fused under `scan`, the cost-weighted update, and the executed-state retraction) takes ≈ 4 ms at best and 10–14 ms sustained under clock throttling—comfortably inside the 50 ms budget of the 20 Hz control rate (Fig. 4)—versus ≈ 200 ms for the vectorized-NUMPY CPU reference of the same K and H . The rollout accounts for $\sim 80\%$ of the step and the executed-state retraction for under 1 ms (2–3 Gauss–Newton iterations). Retracting every rollout state instead (the per-sample-

retraction ablation, Eq. (12)) reaches the same executed $\|c\|$ at $\approx 3\times$ the rollout cost, the $(1 + N_{GN})$ factor predicted by Proposition 2. [TODO: measure the per-sample-retraction cost; re-time with the final dual-arm kinematic model.]

E. Hardware

We run the same tray-transport task on two 7-DoF manipulators and measure feasibility at three levels: (i) the *commanded* constraint $\|c\|$, held to machine precision by the controller; (ii) the *encoder-measured* $\|c\|$, bounded by the servos’ tracking error rather than by the controller—a distinction we make

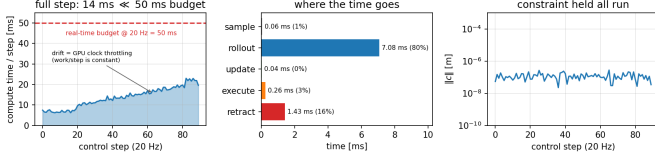


Fig. 4. **Real-time GPU control** (dual-arm, $K = 240$, $H = 18$, RTX 4060). Left: per-step compute time over a transport run, well under the 50 ms budget at the 20 Hz control rate. Middle: the projected rollout dominates the step. Right: the constraint is held throughout. [TODO: re-render with the final controller settings.]

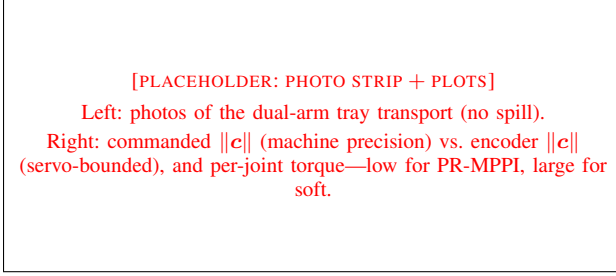


Fig. 5. **Hardware.** [TODO: dual-arm tray transport: commanded vs. encoder $\|c\|$, no spill, per-joint torque comparison.]

explicit; and (iii) physical evidence: a water-filled glass on the tray does not spill, and per-joint torques stay low, whereas the soft baseline builds up larger internal wrenches; for safety the baseline is run only briefly and in a compliant mode. [TODO: run the hardware experiment; numbers, photos, torque traces; confirm F/T or joint-torque sensing.]

V. CONCLUSION

We presented Projection-Retraction MPPI, a sampling-based controller that keeps a manipulator exactly on a hard kinematic equality constraint with two analytic operators: a null-space projection that confines the sampling to the constraint's tangent space, and a retraction that returns the executed configuration to the constraint manifold at each step. Handling feasibility through the geometry rather than the cost decouples constraint satisfaction from task optimization: the projected proposal is a consistent importance sampler for the constrained problem (Lemma 1), the off-manifold drift is $O(\Delta t^2)$ per step and removed by the retraction (Proposition 1), which need touch only the executed state (Proposition 2), and the same construction stays bounded at singularities through a truncated inverse and respects joint limits through a null-space avoidance drift. On a 14-DoF dual-arm tray task the constraint is held to $\sim 10^{-10}$ where soft-penalty and projection-plus-gain [1] baselines are not, the controller stops on the constraint when the commanded task is infeasible, and a GPU implementation runs in real time; the hardware demonstration confirms the same behaviour on two manipulators.

The method targets hard *equality* constraints; inequalities such as collision avoidance remain in the cost, where sampling is strongest, and exactness along a direction is necessarily relinquished once that direction becomes singular. Natural extensions are prioritized stacks of several constraints, inequalities enforced structurally rather than through the cost, and

constraints posed on the full second-order dynamics rather than the kinematic model. Because the two operators need only a differentiable constraint and no learned model, PR-MPPI applies directly to any task whose feasible set is an analytic manifold $c(q) = 0$.

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